Dept. of Electronics and Electrical Communications Engineering

Design & Simulation Laboratory-II [EC69212]



EXPERIMENT – 4

Age Scheduling in Networks

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Date of Submission:

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Introduction

Real-time status updates play a critical role in enabling modern applications such as intelligent transportation systems, smart homes, and environmental monitoring. These systems rely on time-stamped messages generated by sensors or devices that reflect the current state of a physical process. For instance, temperature sensors in forests can provide updates that help predict fires, while cars may exchange velocity and acceleration data to support safe driving decisions. In all these scenarios, the freshness of the information—referred to as the *age of information*—is vital.

The *age of information* at a monitor is defined as the time elapsed since the generation of the latest received packet or status update. Maintaining a low average age is essential for the monitor to make timely decisions. However, ensuring minimal age is a complex challenge due to the trade-offs involved. Simply increasing the update rate can congest the system, causing delays and defeating the purpose of frequent updates. On the other hand, reducing the update rate too much leads to stale information at the monitor.

This experiment explores a simple fundamental question : **Effect of Queue Scheduling policy on Age at Monitor.** We consider queue-based communication systems where packets experience non-zero and random arrival (from a source) and service times(by the server) reaching to a monitor. Our focus is using queues scheduling with different policies with arrival and service times that may be independent and identically distributed (i.i.d.), as in M/M/1/B queueing systems and compare these policies with Age of Information as a metric . We will generate Age plots under different scheduling policies with different load on the server (i.e $\rho = \frac{\lambda}{\mu}$). Insights from this experiments will help you understanding the importance of Freshness of information under given constraints which is crucial for designing efficient real-time monitoring systems in practical, delay-sensitive environments.

Problem Statement

Primary Tasks:

1. Model a Network as a Queue-Server in **matlab** where there a Source generates and sends packets and the server provides service to these packets as a random process (exponentially distributed).
2. Find Average Age at the monitor under the queue Scheduling policies:
   a) FCFS
   b) LCFS
   c) Random Scheduling

   and plot the Age vs Load plot for each scheduling policy. and comment about your observations in each plots and their comparison.

Secondary Tasks:

1. Do Queue scheduling for Round Robin.
2. Generate a Sawtooth Age plot (under FCFS).

Theory

Definition

Age of Information (AoI) is a performance metric that describe the timeliness of a monitor's knowledge of an entity or process. AoI is an end-to-end metric that can be used to characterize latency in status updating systems and applications. consider the given figure below



An update packet with timestamp u is said to have age $t - u$ at time $t \geq u$. An update is said to be *fresh* when its timestamp is the current time t and its age is zero. When the monitor's freshest received update at time t has timestamp $u(t)$, the *age* is the random process defined as:

$$\Delta(t) = t - u(t).$$

In simple words AOI is a metric that captures and quantify the freshness of information in a Communication network.

Time Average AOI

Suppose we have a source–monitor pair. Initially, the Age of Information (AoI) at the monitor increases linearly with time in the absence of any update packets. When an update packet is received at the monitor, the age drops to a smaller value. Consider an update packet i generated at time t_i and received at the monitor at time t'_i. At the reception time t'_i, the age at the monitor resets to $\Delta_M(t'_i) = t'_i - t_i$ which is equal to the total system time of the packet (i.e., the sum of its waiting time in the queue and its service time). At any time t, the age at the monitor is defined as $\Delta_M(t) = t - t_i$ where t_i is the generation time of the most recently received update packet. Thus, the age increases linearly with time until a new update is received, at which point it drops to the system time of the received packet. This results in the characteristic sawtooth evolution of the Age of Information as illustrated in Figure 1.

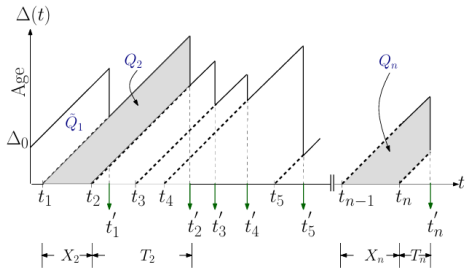


Figure 1: Sawtooth pattern of the age function $\Delta(t)$ over time.

Over an interval $(0, T)$, the *average age* is given by:

$$\Delta_T = \frac{1}{T} \int_0^T \Delta(t)\, dt$$

References

[1] S. Kaul, R. Yates, and M. Gruteser, "Real-time status: How often should one update?" IEEE INFOCOM, 2012.

[2] Yates, Roy D., et al. "Age of information: An introduction and survey." IEEE Journal on Selected Areas in Communications(2021)

[3] Kaswan, Priyanka, et al. "Age of Information in Gossip Networks: A Friendly Introduction and Literature Survey Invited Paper." IEEE Transactions on Communications (2025).